

Qwestor Hardcore Research:

Rough requirements for guiding a vibe-coded prototype

WORK IN PROGRESS

The document presents rough requirements intended to guide a fast OpenClaw-based prototype of the Hardcore Research pipeline, which will be a crucial and differentiating component of Qwestor. It does not cover a full product specification.

This prototype should be considered a baseline of what OpenClaw with reasonable and fast modifications can do today, which we will improve in our MVP.

Particular skills and details should be defined during the process of implementation.

Prototype features:

- Integrated search capabilities.
 - Uses APIs to search for sources (to begin, only web and academic papers).
- Short-term and Long-term memory division
- Conversational interface:
 - Allows the user to enter prompts in the form of text, attached files.
- Hardcore research:
 - An interactive pipeline to assist users to advance research.
 - Capable of co-creating and executing a research idea and plan.
 - Can write and execute code, analyze results, and take action on them.
 - Interactive loops to advance research, until human input is needed.
- Long-term Persistent memory:
 - Can refer to older user dialogues and operate with up-to-date information.
- Report generation:
 - Can generate clearly structured reports based on user requests and templates.
- Credibility and Safety:
 - Verifies that obtained sources are valid.
 - No hallucinations, based only on clear facts.
 - Avoids destructive actions in the user's system (to the extent possible).

x`Flow of the prototype's work (Based on [Finance usecase](#))

1. Dialogue communication, problem identification

2. Creating a research topic, creating a folder for a project (local or cloud storage).
 3. Creating a research plan
 4. Executing a step-by-step plan (with validation of each step with a plan)
 5. Writing each step to the resulting article (in a folder (step 2))
 6. Drawing conclusions, writing to a file
 7. Validation of the global results obtained, conclusions on the need for a new research
- (Validation with the User at each step only if necessary)

Flow steps could be reasonably modified, with a previous discussion about it.

FOR LATER, MORE GENERAL MVP REQUIREMENTS

Qwestor Rough MVP draft

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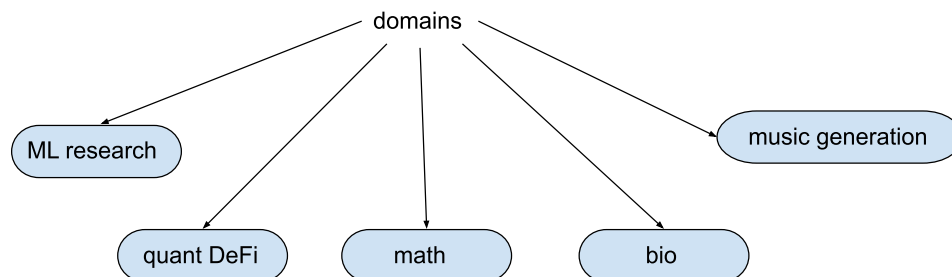
Qwestor's idea / goal: a project-based research assistant for advancing structured research workflows.

Differentiating features:

- Hardcore Research pipeline.
- Project-oriented user interface.
- Configurable autonomy.
- Structured, persistent memory with reliable context recall.

Principles:

- ☐ All features start with basic LLM-functionality, then evolve to integrate Hyperon modules.
- ☐ LLM-modules should be replaceable for Hyperon ones as they are ready.
- ☐ Self-hosted product.



MVP rough features across 5 domains above (in short description):

- Project-centric UX.
- Integrated search capabilities.
- Conversational interface.
- Short-term and Long-term memory.
- Hardcore research capabilities.
- Autonomy.
- Report generation.
- Credibility and Safety.
- Personalization and Adaptive Responses.
- Interaction with audio and media files for their processing / generation.
- *External Platform Interoperability Module (e.g., Rejuve.Bio, music platforms etc.)?*

MVP requirements for **ML experimentation / research / paper-writing** domain:

- Project-centric UX:
 - Projects dashboard
 - Projects are (optionally) defined with goals, open questions, hypothesis, tasks, research results, conversations.
 - Uncategorized interactions (like Global cross-project chat) also exist, and can migrate to a specific project when convenient.
- Conversational interface
 - Allows the user to enter prompts in the form of text, attached files, or audio data.
- Hardcore research capabilities:
 - An interactive pipeline to assist users to advance research.
 - Capable of co-creating and executing a research idea and plan.
 - Can write and execute code, analyze results, and take action on them.
 - Interactive loops to advance research, until human input is needed.
 - Can spawn agents to run different tasks in parallel, if suitable.
 - Can continue with well-defined tasks while it waits for the user's input for an undefined task.
- Autonomy:
 - Can operate without human intervention when convenient/motivated.
 - Can execute scheduled tasks without direct user participation.
 - Can spawn agents to run different tasks in parallel, if suitable.
 - Can continue with scheduled, well-defined tasks while it waits for the user's input for an undefined task.

Motivation

- Short-term memory:
 - Quick Q-A sessions that do not require storage in memory.
- Long-term Persistent memory:
 - Can refer to older user dialogues and operate with up-to-date information.
 - No hallucinations, based only on clear facts.
- Report generation:
 - Can generate clearly structured reports based on user requests and templates.
 - Reports can be imported into third-party sources (such as Google Docs) or in various formats (docx, pdf, pptx, xlsx and etc.).
- Integrated cross-projects search capabilities.
- Autonomous Discovery.
- Credibility and Safety.
- Personalization and Adaptive Responses.

Hyperon and(or) Claw-like modules integration and their contribution across 5 domains above:

- **OpenPSI integration:**
 - Autonomous selection of the optimal action from a set of available options.
 - Deep contextual adaptability - taking into account the user's level of training, responding to complexity, caution after making a mistake when giving an answer.
 - Maintaining consistency of behavior - interaction style develops gradually based on the history of dialogues, rather than being reset with each new request.
- **PLN integration:**
 - Building a knowledge base based on facts provided by the user.
 - Logical chaining and inference based on natural language queries.
- **Claw-like modules integration:**
 - A set of tools for direct interaction with the OS and external services (file system operations, browser control, code execution, network operations, shell commands).
 - The ability to switch between proprietary and open models by simply changing the configuration, without rewriting the code.
 - Built-in integration with messengers.
 - Defining access rights, contextual triggers, and execution conditions to ensure complete security and transparency of the assistant's actions.

Conclusion - Best approach for the Qwestor HR MVP in the focus domain:

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- Autonomy
 - Can operate without human intervention when convenient/motivated.
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- Persistent memory
- Integrated search capabilities
- Autonomous Discovery
- Report generation
- Conversational interface
- Credibility and Safety
- Personalization and Adaptive Responses